

**MAHATMA JYOTIBA PHULE TELANGANA
BACKWARD CLASSES WELFARE RESIDENTIAL
DEGREE COLLEGE(W)**

STATION GHANPUR

Group : B.Sc (M.S.Cs) – I

INTERNAL - I EXAMINATION QUESTION PAPER

Semester: I Subject: Basic Statistics & probability

Max Time: $1\frac{1}{2}$ Hr. Max Marks: 20

ANSWER ALL THE QUESTIONS:

10x2 = 20M

- 1) Define geometric mean?
- 2) Define coefficient of variation?
- 3) Calculate Median for the following data

26, 8, 6, 12, 15, 32

- 4) Derive the limits of Bowley's coefficient of skewness?
- 5) Define kurtosis?
- 6) Define Central and Non Central moments?
- 7) Find Mode to the following frequency distribution

X	1	2	3	4	5	6	7	8
f	4	9	16	25	22	15	7	3

- 8) Define various types of skewness?
- 9) Find Quartile Deviation of the values
2, 4, 10, 8, 6, 12, 5
- 10) State the properties of Arithmetic mean?

Faculty of Statistics

Examination Branch

Principal



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STATION GHANPUR

Class : B.Sc (M.S.Cs) – II

INTERNAL - I EXAMINATION QUESTION PAPER

Semester: III Subject: (Statistical methods & Theory of estimation)

Max Time: $1\frac{1}{2}$ Hr. Max Marks: 20

ANSWER ALL THE QUESTIONS:

10x2 = 20M

- 1) What are the differences between correlation & regression?
- 2) Define positive correlation?
- 3) If X and Y are random variables and a, b, c & d are any numbers provided only $a \neq 0$, $c \neq 0$ then show that $r(ax + b, cy + d) = \frac{ac}{|ac|} r(x, y)$
- 4) Define regression analysis?
- 5) Write the properties of correlation coefficient?
- 6) Define regression coefficients?
- 7) Prove that the geometric mean b/w regression coefficients is the correlation coefficient?
- 8) Define multiple correlation?
- 9) Define attributes with examples?
- 10) What is meant by the order of class?

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STATION GHANPUR

**Group : B.Sc (M.S.Cs, M.P.Cs, B.Z.C) – II
INTERNAL - I EXAMINATION QUESTION PAPER**

Semester: III Subject: Bio Statistics Max Time: IIIr. Max Marks: 20

ANSWER ALL THE QUESTIONS:

2x5 = 10M

- 1) What is meant by Range, Quartile deviation, Mean deviation, Standard deviation ?
- 2) Find mode to the following data

Class interval	0-10	10-20	20-30	30-40	40-50	50-60
f	2	7	16	25	13	5


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MJPTBCWRDC(W), STATION GHANPUR LOCATED AT PEMBARTHY
INTERNAL EXAMINATION – I
ACADEMIC YEAR 2026-2027

Group: B.Sc (MSCs) Subject: Statistics Year: II

Semester: III

Marks: 20

Answer all the following questions.

10×2=20

- 1) Define consistency.
- 2) If T is an unbiased estimator of θ , show that T^2 is not an unbiased estimator for θ^2 .
- 3) Define minimum variance unbiased estimator.
- 4) Explain method of moments.
- 5) Obtain MLE for λ in exponential distribution.
- 6) Find sufficient statistic for θ for the following density function. $f(x, \theta) = \theta x^{\theta-1}, 0 < x < 1, \theta > 0$.
- 7) Find normal equations for second degree parabola using principle of least squares.
- 8) Obtain 95% confidence interval for μ in the normal population, where σ is known.
- 9) Define interval estimation.
- 10) State Cramer Rao inequality.

MJPTBCWRDC(W), STATION GHANPUR LOCATED AT PEMBARTHY
INTERNAL EXAMINATION – I
ACADEMIC YEAR 2026-2027

Group: B.Sc(MSCs)

Subject: Statistics

Year: III

Semester: V

Marks: 20

Answer all the following questions.

10×2=20

- 1) Define Standard Error.
- 2) Define simple random sampling without replacement.
- 3) Sample mean is an unbiased estimator of population mean in SRSWOR. i.e, $E(\overline{y_n}) = \overline{Y_N}$
- 4) Difference between SRSWOR and SRSWR.
- 5) In SRSWR sample mean square is an unbiased estimator of the population variance. i.e, $E(s^2) = \sigma^2$.
- 6) Define stratified random sampling.
- 7) Define optimum allocation.
- 8) Define cost function.
- 9) Define systematic sampling.
- 10) Define types of sampling.

